Lab 3.1: Pointer Manipulation

lab3.1.go

```go
package main

import "fmt"

type Student struct {
    Name  string
    Age   int
    Marks float64
}

func increment(n *int) {
    *n = *n + 10
}

func main() {
    fmt.Println("=== Address and value ===")
    x := 42
    p := &x
    fmt.Println("Value of x:", x)
    fmt.Println("Address of x (&x):", &x)
    fmt.Println("Pointer p:", p)
    fmt.Println("Value via pointer (*p):", *p)

    *p = 100
    fmt.Println("After *p = 100, x =", x)

    fmt.Println("\n=== Pass by reference ===")
    num := 5
    fmt.Println("Before function call:", num)
    increment(&num)
    fmt.Println("After function call:", num)

    fmt.Println("\n=== Struct with new() ===")
    s := new(Student)
    s.Name = "Asha"
    s.Age = 20
    s.Marks = 87.5
    fmt.Printf("Initial: %+v\n", *s)

    s.Marks = 91.0
    s.Age++
    fmt.Printf("Updated: %+v\n", *s)
    fmt.Printf("Name via pointer: %s | Address of struct: %p\n", s.Name, s)
}
```

Lab 3.1: Output

